

- 1. Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.**

Steps Performed

- Shut down the computer completely.
- Unplugged the power cable and disconnected all external devices.
- Pressed and held the power button for about 10 seconds to discharge any remaining power.
- Opened the computer case (or laptop service panel) using a screwdriver.
- Located the RAM slots on the motherboard.
- Gently pushed the retaining clips on both sides of the RAM module outward.
- Carefully removed the RAM module by holding it from the edges without touching the gold contacts.
- Inspected the RAM slot and module for dust or visible damage.
- Aligned the notch on the RAM module with the slot.
- Firmly pressed the RAM module back into the slot until both retaining clips clicked into place.
- Closed the computer case and reconnected the power cable and peripherals.
- Powered on the computer and verified that it booted normally.

Issues Observed

- No physical damage was observed on the RAM module or slot.
- The RAM module required firm, even pressure before the retaining clips locked into place.
- After reseating, the computer started successfully and detected the installed RAM correctly.

- 2. Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.**

Step	What to Check	Expected Result / Action
1. Identify the Symptoms	Check for no display, screen flickering, graphical artifacts, game crashes, or system freezes.	Note the exact symptoms and when they occur (boot, gaming, or normal use).
2. Inspect Physical Connections	Turn off the PC and check that the GPU is firmly seated in the PCIe slot. Ensure the PCIe power cables are securely connected.	Reseat the GPU and reconnect power cables if necessary.
3. Check Monitor and Display Cable	Verify that the monitor is powered on and connected to the GPU (not the motherboard). Test with another HDMI/DisplayPort cable or monitor.	Replace any faulty cable or monitor if needed.
4. Test Software and Drivers	Boot into Windows, open Device Manager, and check if the GPU is detected. Update or reinstall the latest graphics driver from the GPU manufacturer.	The GPU should be detected without errors, and driver issues should be resolved.
5. Monitor GPU Temperature	Use a hardware monitoring tool to check GPU temperatures while gaming or under load.	Normal temperatures are generally below 85°C . Clean dust or improve cooling if overheating occurs.
6. Test the GPU in Another System (or Test Another GPU)	Install the GPU in a known working PC, or install a working GPU in the current PC.	This helps determine whether the issue is with the GPU or another component (motherboard, PSU, etc.).
7. Confirm the Fix	Restart the PC, play a graphics-intensive game, or run a GPU stress test for 15–30 minutes.	If there are no crashes, artifacts, or display issues, the GPU is functioning correctly and the problem has been resolved.

- 3. Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.**

Scenario A: Unplugging the Boot Drive (System Powered Off)

If you power down the system, remove the SATA or NVMe drive containing your operating system (OS), and turn the PC back on, here is what happens:

What You See Before Reconnecting:

- **On-Screen Errors:** The system will fail to load the OS. You will typically see a black screen with white text displaying one of the following classic motherboard errors:
 - Reboot and Select proper Boot device or Insert Boot Media in selected Boot device and press a key
 - No bootable device found
 - Operating System not found
- **BIOS/UEFI Indicators:** If you press the key to enter the BIOS (usually F2, Del, or F12), the Boot Priority list will be empty or will only show alternative network boot options (like PXE IPv4/IPv6). In the "Storage Information" or "NVMe Configuration" tab, the specific M.2 slot or SATA port will read **"Empty"** or **"Not Present"**.

What You See After Reconnecting:

- Once you power down, reseat the drive securely, and power back on, the motherboard will detect the hardware during POST (Power-On Self-Test).
- The system will successfully find the Windows Boot Manager (or GRUB for Linux) and boot normally to your lock screen.
- In the BIOS, the drive's model number (e.g., *Samsung SSD 980 PRO* or *WD Blue SN570*) will reappear in the boot order list and storage registry.

Scenario B: Hot-Swapping a Secondary Drive (System Powered On)

"Hot-swapping" means unplugging a drive while the PC is running. *Note: SATA supports hot-swapping if AHCI is enabled in the BIOS. NVMe hot-swapping is supported by the NVMe protocol but requires specific motherboard PCIe hot-plug support, which is rare on consumer desktop boards.*

What You See When Unplugging (The "Surprise Removal"):

- **OS Level:** The drive will instantly vanish from Windows File Explorer or Linux File Manager.
- **Error Messages:** If any files or programs were open on that drive, they will immediately crash. You will see Windows pop-up errors like:
 - A device which does not exist was specified.
 - Location is not available: D:\ is unavailable.
- **System Logs:** If you open the Windows Event Viewer and check the System logs, you will see a Warning/Error entry (Event ID 157) stating: Disk X has been surprise removed.

What You See When Reconnecting:

- **OS Level:** You will hear the standard Windows "Device Connected" chime. The OS will automatically scan the bus, detect the file system, and mount the drive.

- **Indicators:** The drive letter will pop back up in File Explorer. If the surprise removal caused file system corruption, Windows might immediately prompt a notification: There's a problem with this drive. Scan the drive now and fix it.
4. **Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.**

Component to Check First: RAM

Reasoning:

- Faulty RAM can cause random restarts, crashes, and system instability.
- Loose RAM modules may interrupt communication with the motherboard.
- RAM issues are relatively easy to test and rule out before checking more complex hardware.

Diagnostic Steps

1. Power Off the PC

- Shut down the computer.
- Disconnect the power cable and press the power button for a few seconds to discharge remaining power.

2. Reseat the RAM

- Open the PC case.
- Remove the RAM module(s) carefully.
- Clean the contacts gently if needed and reinstall them firmly until the locking clips click into place.

3. Test One RAM Stick at a Time

- If multiple RAM modules are installed, boot the PC with only one stick installed.
- Repeat with each RAM stick and in different motherboard slots to identify a faulty module or slot.

4. Run a Memory Diagnostic

- Use **Windows Memory Diagnostic** or **MemTest86** to scan for memory errors.
- If errors are reported, the RAM may need to be replaced.

5. Check BIOS Settings

- Ensure RAM speed and voltage are set to the manufacturer's recommended values.
- Disable any unstable overclocking or XMP/EXPO profile temporarily to see if stability improves.

6. Monitor System Stability

- Restart the PC and use it normally or run a stress test.
- If the random restarts stop, the issue was likely caused by the RAM or its configuration.